

- Maps Maps show up as rectangular icons.
- Materials Materials show up as rendered spheres.

3.2 Creating and Editing Materials

Creating a material requires several steps. First, you need to decide the type of material to use. 3ds Max has many material types, which cover a wide range of surface types and options. After the basic material is selected, you can add textures and change parameters to fine-tune the material. Each material has its own characteristics, depending on the type of surface and style of rendering. Materials can be added via plug-ins, so the number of different materials you may encounter can be quite robust. This section describes a few of the more common materials used within 3ds Max.

Standard Material

The standard material is the default material used in 3ds Max. It can simulate a wide variety of different surfaces by using different shading types. The shading types are selected by using a pull-down menu, and each type simulates a different type of surface and has its own set of parameters.

Anisotropic simulates surfaces that have directional highlights. Good for such materials as brushed metal, silk and hair.

Blinn: A versatile shading algorithm and probably the most popular among artists. Has fairly soft highlights by default.

Metal: A shader that resembles a metallic surface. Metal shading has a distinct curve for specular highlights. Metal surfaces also have glancing highlights. Metal materials calculate their own specular colour based on the diffuse colour and the colour of the light, so you can't set it.

Multi-Layer: A material that has two sets of anisotropic specular highlights. Good for creating surfaces that are highly polished or have depth, such as automobile paint.

Shader: The standard shader has a basic parameters rollout. This allows you to change the shading type as well as four other parameters. Wire renders the material in wireframe, 2-Sided renders the back side of a surface, Face Map maps a texture once per face of an object, and Faceted removes smooth shading calculations.

Oren-Nayar Blinn: A softer version of the Blinn shader, this is good for non reflective surfaces such as cloth.

Phong: One of the oldest shading algorithms, this creates a surface with distinct specular highlights. Supports reflections and is good for creating shiny artificial surfaces such as plastics and some glass.